



OnePlus TV to launch soon

The OnePlus TV will reportedly launch soon in India. The company first announced plans to launch a TV last year. <https://digit.in/july19-14>



No SD card for Note 10?

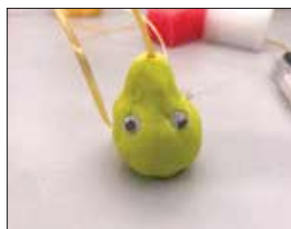
According to some leaks, the Samsung Galaxy Note 10 may not have the much-loved SD card support this time. <https://digit.in/july19-15>



is not lopsided. While you can cover the bottom as well, we do not recommend this. Be sure to wash your hands or use a sanitiser after handling the clay.

STEP 9

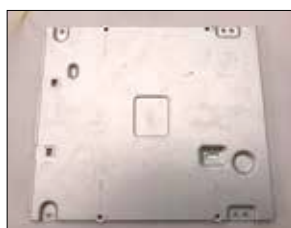
Now you can embellish the clay. We went for a lumpy alien being by adding googly eyes. These can be purchased from a craft or stationery store. We positioned four googly eyes



in a circle. During the chaotic movements of the pendulum, the googly eyes also jump around, which introduces another chaotic movement to the mix.

STEP 10

Now we need a metallic plate. The plate can be big or small, but the larger the surface area, the better. We used the base of an old DVD player, but anything that magnets stick to can be



used. A base of any material can be used if the magnets are stuck to the base with glue, but a metal base is better as it allows you to rearrange the magnets, which ultimately can let you tweak the duration for which the pendulum keeps moving chaotically.

STEP 11

Arrange the remaining magnets in a random array on the metallic plate. You can use any pattern that you want, and as many magnets as you want. A minimum of three magnets are recommended. Make sure that the pole facing upwards repels the pole at the bottom of the pendulum. One or two mag-



nets in the mix can attract the pendulum, provided they are surrounded by magnets that repel the pendulum.

STEP 12

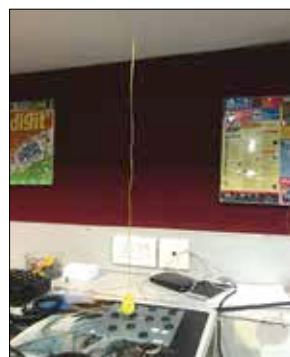
The string has to be attached to a stand of sorts. We are using an overhanging shelf over the desk. The string is just stuck to the shelf using tape. Note that this approach allows the length of the string to be adjusted after it is stuck, allowing us to fine



tune the motion of the pendulum. Any excess string can be cut off after it has been firmly attached.

STEP 13

This is the finished pendulum. When you swing it,



there is no regular back and forth motion. Instead, the pendulum gets deflected in unpredictable ways by the magnets on the plate, and it keeps jumping all over the place erratically. Now, the arrangements of the magnets, the number of magnets, and the length of the string can be adjusted, to increase the duration for which the pendulum keeps moving once it is swung.

STEP 14

Give yourself a pat on your back and enjoy your brand

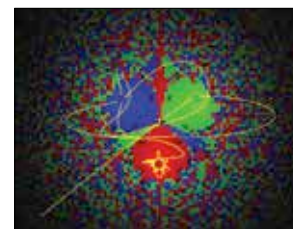


new desk toy. The chaos pendulum makes for great slow motion videos. You can also add small figures of people, the kind that are used for architectural models, to see which ones survive and

which ones are toppled by the pendulum alien.

THE PHYSICS

Every time the pendulum swings, it gets deflected by various magnets, which can be imagined as diffused spheres of magnetism. Eventually, the pendulum settles in a particular position. The movement of the pendulum is completely dependent on the initial positions. However, even slight variations in the initial position can drastically change the trajectory. Chaos theory is used to study such systems, where a small change in an initial condition can result in big differences over periods of time. The behavior of the magnetic chaos



pendulum is the butterfly effect in action.

Although the motion of the pendulum looks erratic, there is a way to actually predict how it will behave. Essentially, this involves calculating the total acceleration of the pendulum. This is the acceleration caused by the pullback of the pendulum, the acceleration caused by each of the individual magnets it encounters, and the deceleration because of air resistance. If you want to calculate the motion of the pendulum based on all the forces acting on it, and also see a simulation that performs the calculations, head over to <https://digit.in/chaospendulum>.